

Section 3: Scientific Force Multiplication (C3)

Assumption Used

Assumption 1 (Documented Scientific Breakthroughs (A5)). *The following AI-driven scientific results are documented facts:*

1. **AlphaFold2** (DeepMind, 2021): *Solved the protein-structure prediction problem to near-experimental accuracy. A5 explicitly states this problem had been open for 50 years despite active human research. The resulting database covers >200M protein structures.*
2. **AI drug discovery** (Insilico Medicine, 2023): *An AI-designed drug candidate reached IND approval in **18 months**. The historical human-only average is **4–6 years** (48–72 months).*
3. **GNoME** (DeepMind, 2023): *Discovered **2.2 million** novel stable crystal structures, surpassing the total of all prior human experimental discovery (approximately 20,000 structures) by a stated ratio of > 100×. The structures are described as novel (not previously in the human record).*
4. **Lean4 AI proof assistance** (2023–2024): *AI-assisted proof search produced novel proof strategies for open combinatorics problems; such tasks previously required years of specialist effort.*

Definitions

Definition 1 (Prior Human Record). *The prior human record in a domain is the collection of documented scientific results produced by human researchers before the AI system in question was deployed, as described in A5.*

Definition 2 (Beyond-Record Contribution). *An AI result R is a beyond-record contribution in domain D if R is explicitly described in A5 as either: (a) a solution to a problem stated in A5 to have been open (unsolved) in the prior human record, or (b) a collection of results whose scale or novelty is stated in A5 to exceed the entire prior human record in D .*

Definition 3 (Time-to-Discovery Speedup). *For a discovery task T , the time-to-discovery speedup is the ratio $\tau_{\text{human}}/\tau_{\text{AI}}$, where τ_{human} is the documented human-only time and τ_{AI} is the documented AI-assisted time for the same task, both as stated in A5.*

Definition 4 (Independent Scientific Problem Areas). *Problem areas are independent if each is a distinct recognised scientific discipline and solving one does not constitute a solution to any other. The four areas named in A5 — protein-structure prediction (structural biology), drug candidate design (pharmacology/medicine), stable crystal-structure discovery (materials science), and formal combinatorics proof search (mathematics) — are stipulated as independent by this definition: each belongs to a different branch of science, and no prior-record solution in one area is a solution in another.*

Statement to Prove (C3)

Theorem 1 (Scientific Force Multiplication). *Under Assumption 1 and Definitions ??–??, AI systems constitute a cross-domain scientific force multiplier:*

- (i) *In at least two independent scientific domains, AI has made beyond-record contributions (Definition ??).*
- (ii) *A5 documents at least one time-to-discovery speedup $\geq 2\times$, and at least one beyond-record contribution whose scale exceeds the prior human record by $> 100\times$.*
- (iii) *The documented gains span at least four independent scientific problem areas (Definition ??).*

Proof. Step 1: Domain inventory. A5 names four case studies in four distinct problem areas: protein-structure prediction (α), drug candidate design (δ), stable crystal-structure discovery (γ), and formal combinatorics proof search (λ). By Definition ??, these four areas are independent. This immediately establishes claim (iii): the gains span four independent problem areas.

Step 2: Beyond-record contributions in two domains.

Case α (AlphaFold2). A5 states that protein-structure prediction was an *open problem for 50 years*. AlphaFold2 solved it. A5’s own language characterises this as an open problem in the prior human record. Therefore case α satisfies Definition ??(a). Domain: structural biology.

Case γ (GNoME). A5 states GNoME produced 2.2 million novel stable crystal structures and explicitly gives the ratio over the prior human experimental corpus as $> 100\times$. This directly satisfies Definition ??(b): the scale of the result exceeds the entire prior human record by $> 100\times$, as stated in A5. Domain: materials science.

Two independent domains (α and γ) exhibit beyond-record contributions. This establishes claim (i).

Step 3: Quantified acceleration.

Time-to-discovery speedup (case δ). By Definition ??:

$$\tau_{\text{human}} \geq 4 \text{ years} = 48 \text{ months}, \quad \tau_{\text{AI}} = 18 \text{ months (A5)}.$$

Using the conservative (shortest) human baseline from A5:

$$\text{Speedup}_{\delta} = \frac{\tau_{\text{human}}}{\tau_{\text{AI}}} \geq \frac{48}{18} = \frac{8}{3} \approx 2.67 > 2.$$

Hence A5 documents a time-to-discovery speedup $> 2\times$.

Scale multiplication $> 100\times$ (case γ). A5 explicitly states the GNoME discovery-count ratio is $> 100\times$. We adopt this A5-stated ratio directly.

Both quantified thresholds in claim (ii) are established.

Conclusion. Claims (i), (ii), and (iii) are derived exclusively from A5 and Definitions ??–??. Together they establish that AI is a cross-domain scientific force multiplier with at least two beyond-record contributions, one confirmed $> 2\times$ time speedup, one confirmed $> 100\times$ scale multiplication, spanning four independent scientific disciplines. $\square$